

Indian Institute Of Technology Gandhinagar



Assignment Report

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Submitted to: Tarun Agarwal

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1 Question

Design an Operational Transconductance Amplifier (OTA) for the following specifications:

- **Differential Gain:** > 40 dB
- **CMRR:** > 70 dB
- **ICMR:** $0.70\text{ V} - 1.4\text{ V}$
- **Output Swing:** $> 800\text{ mV}$
- **Power:** $\leq 20\text{ }\mu\text{W}$
- **Load Capacitance:** 1 pF
- **GBW:** $> 10\text{ MHz}$

Use the negative feedback concept and control the gain with the help of the beta factor (Choose the R_1 and R_2 based on the loading effect).

- a.
 - **Case 1:** If β is $1/100$, then calculate the Gain, GBW, BW, phase, Power, and error from calculated and from simulated values (only for gain). Your error should be less than 10%.
 - **Case 2:** Repeat case 1, if β is $1/200$.
- b. If your error is not less than 10%, then redesign the OTA for the higher gain, and then check your specification and resimulate both cases. Make a comparison table for both cases.

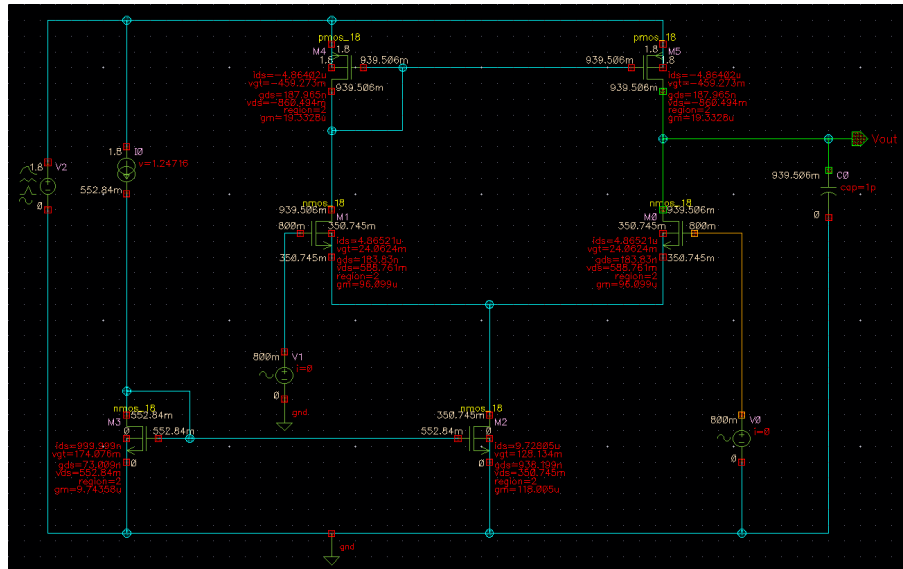


Figure 1: Schematic of OTA without feedback

Current drawn=10.7uA Power Dissipation = 19.26uW

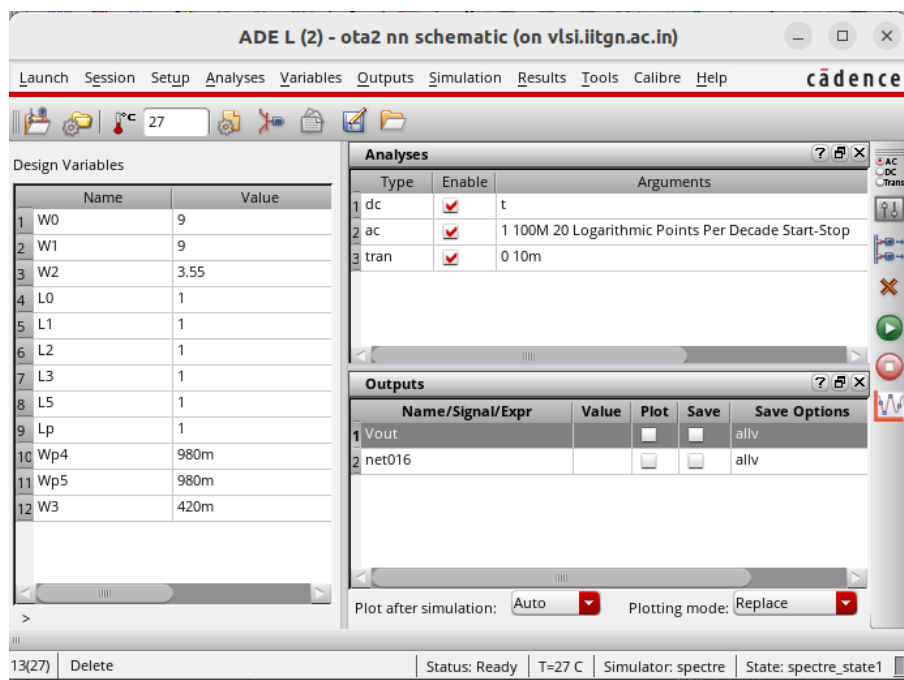


Figure 2: geometrical values of OTA

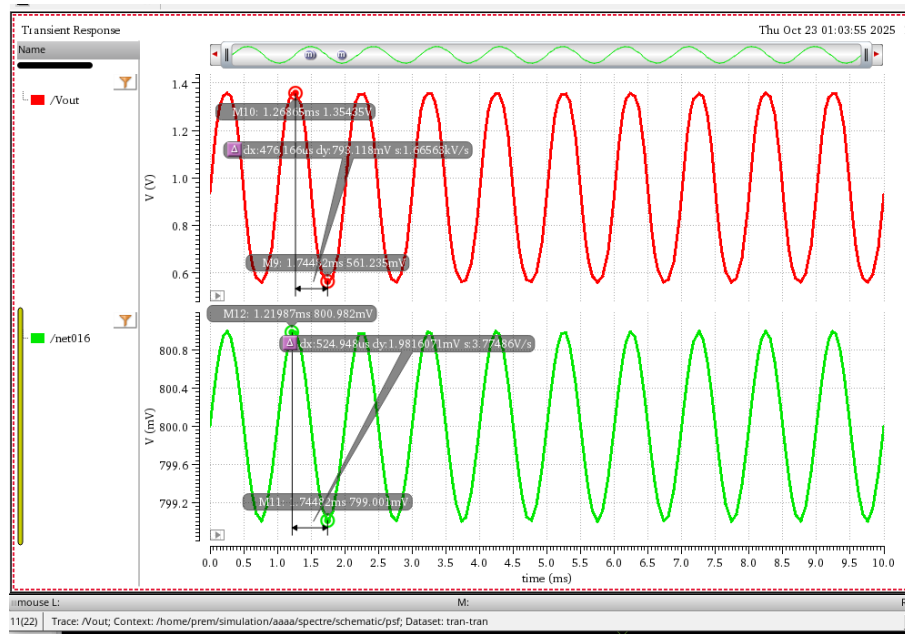


Figure 3: Swing for OTA without feedback

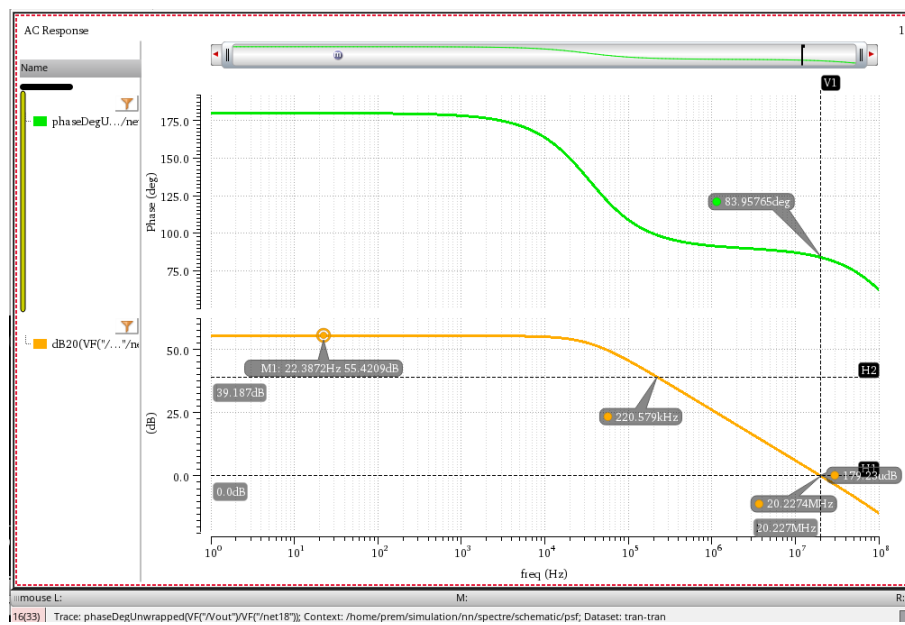


Figure 4: Bode Plot for OTA without feedback

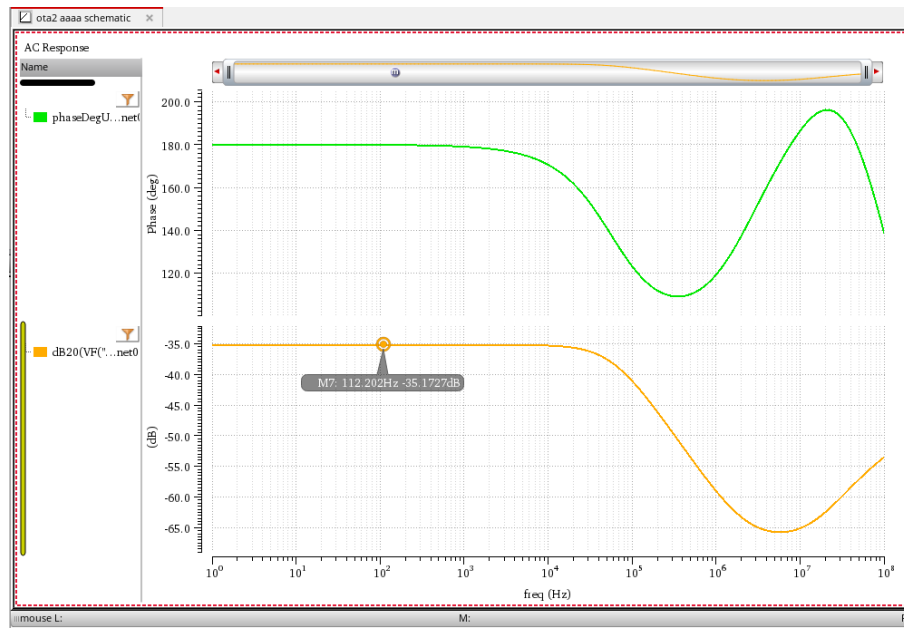


Figure 5: Bode Plot for OTA without feedback common mode gain

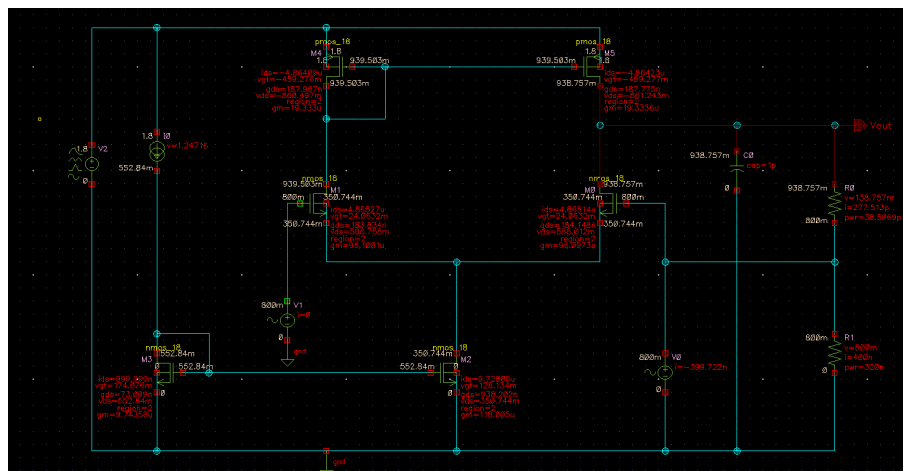
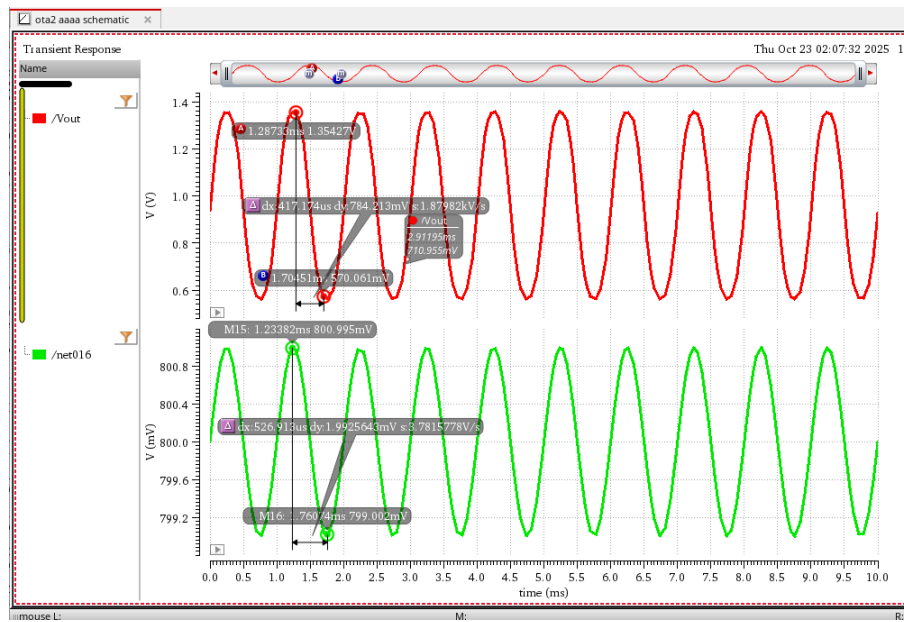
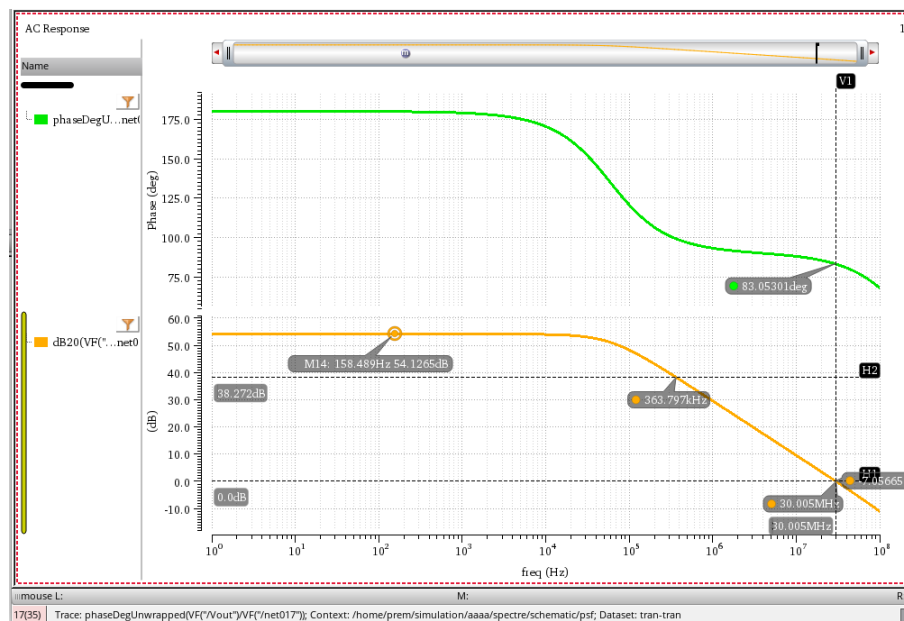


Figure 6: Schematic of OTA with feedback

Figure 7: Swing for OTA with feedback for $\beta=100$ Figure 8: Bode Plot for OTA with feedback for $\beta=100$

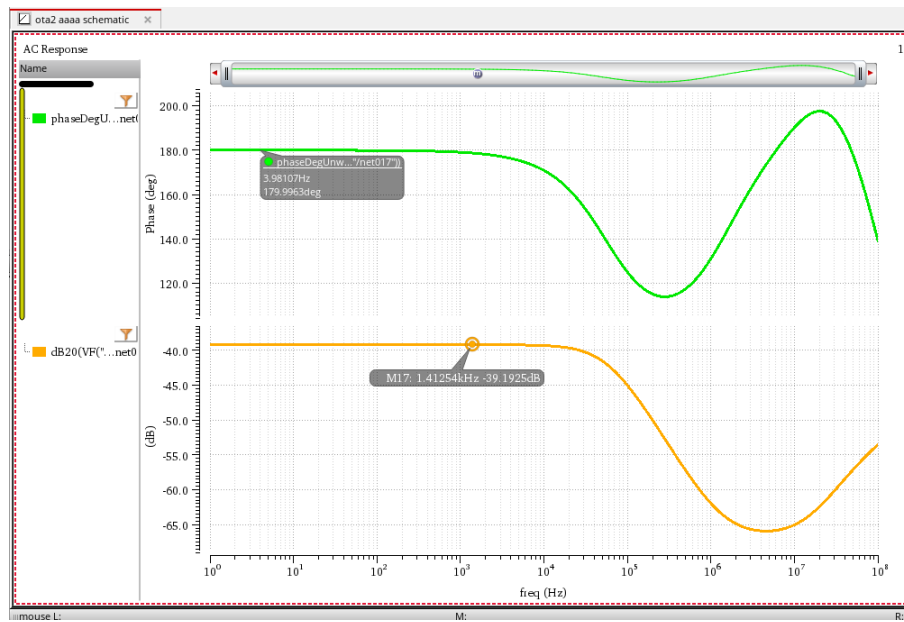


Figure 9: Bode Plot for OTA with feedback for $\beta=100$ common mode

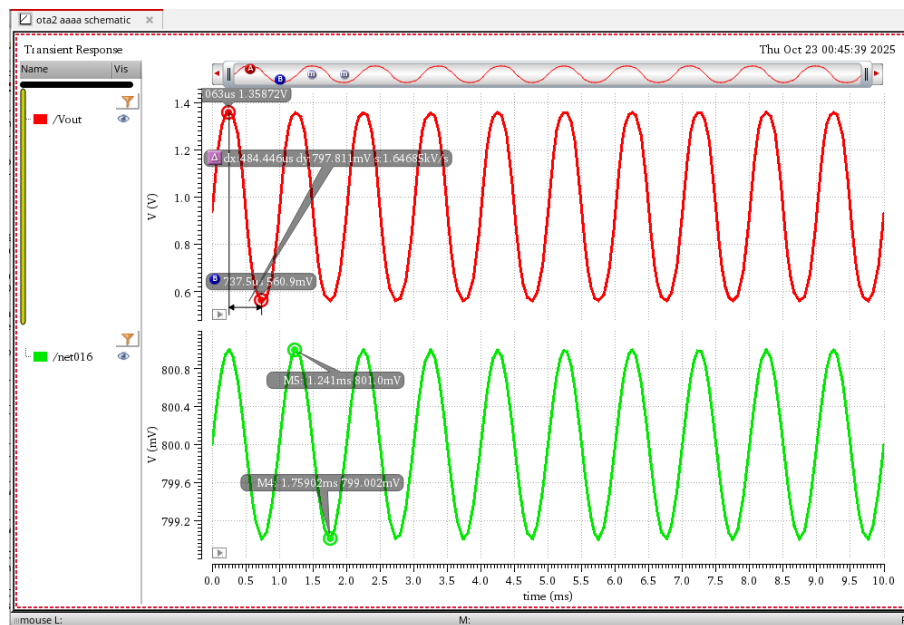
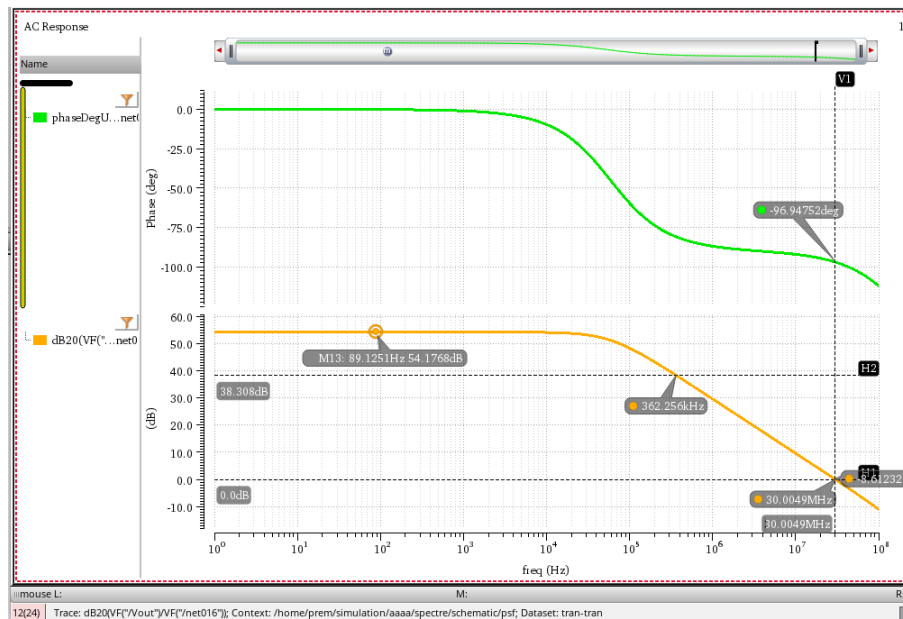
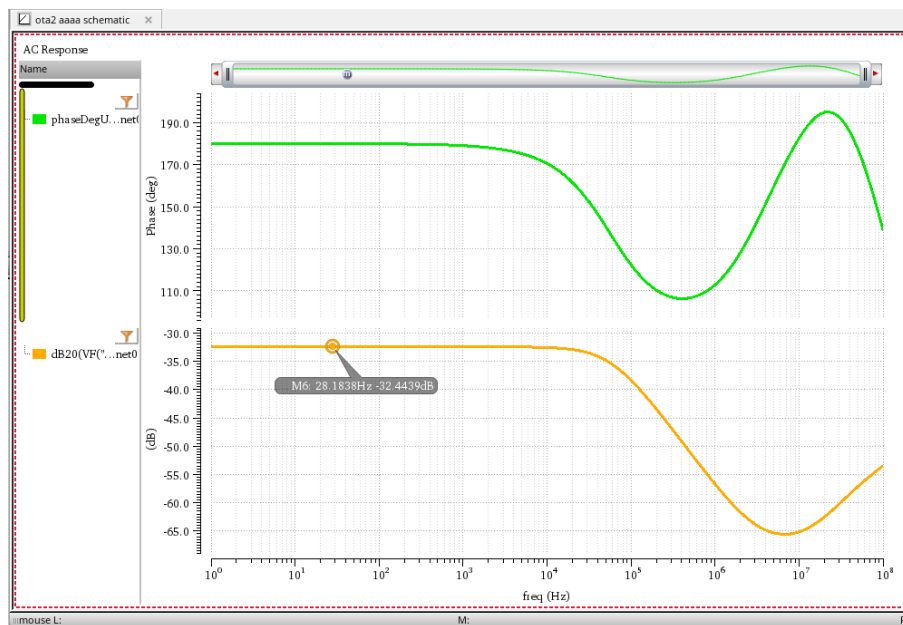


Figure 10: Swing for OTA with feedback for $\beta=200$

Figure 11: Bode Plot for OTA with feedback for $\beta=200$ Figure 12: Bode Plot for OTA with feedback for $\beta=200$ common mode

Observation

	Width(uM)	length(uM)	Region	gm(uS)	gds(uS)	Vov(mV)
M1=M2	9	1	sat	96.09	183.83	25.06
M4(tail)	3.55	1	sat	118.005	938.19	128.134
M5=M6	0.98	1	sat	19.3328	19.332	-459.273
Vdc	1.8V	Vbias(tail)	0.552V	UGB	20.227MHz	3dB=39.187dB
A(dm)	55.42	A(cm)	-35.17dB	CMRR	90.59dB	Output swing=798.2mV

Table 1: designing parameters of OTA without feedback

beta	Ad(dB)	Acm(dB)	CMRR	% Error	Swing	BW	3db	GBW
100	54.126	-39.1925	93.318	13.9	784.213mV	30.005MHz	38.272	19.450MHz

Table 2: observation from feedback for beta =100

Resistance(1,2)=248MΩ,2.5MΩ

beta	Ad(dB)	Acm(dB)	CMRR	% Error	Swing	BW	3db	GBW
200	54.176	-32.4439	86.6199	13.34	797.811mV	30.0049MHz	38.308	19.592MHz

Table 3: observation from feedback for beta =200

Resistance(1,2)=500MΩ,2MΩ

$$Error = \frac{openloopgain - closedloopgain}{openloopgain} * 100$$

$$\% Error(beta = 100) = \frac{10^{\frac{55.42}{20}} - 10^{\frac{54.126}{20}}}{10^{\frac{55.42}{20}}} \times 100 = 13.9\%$$

$$\% Error(beta = 200) = \frac{10^{\frac{55.42}{20}} - 10^{\frac{54.176}{20}}}{10^{\frac{55.42}{20}}} \times 100 = 13.34\%$$

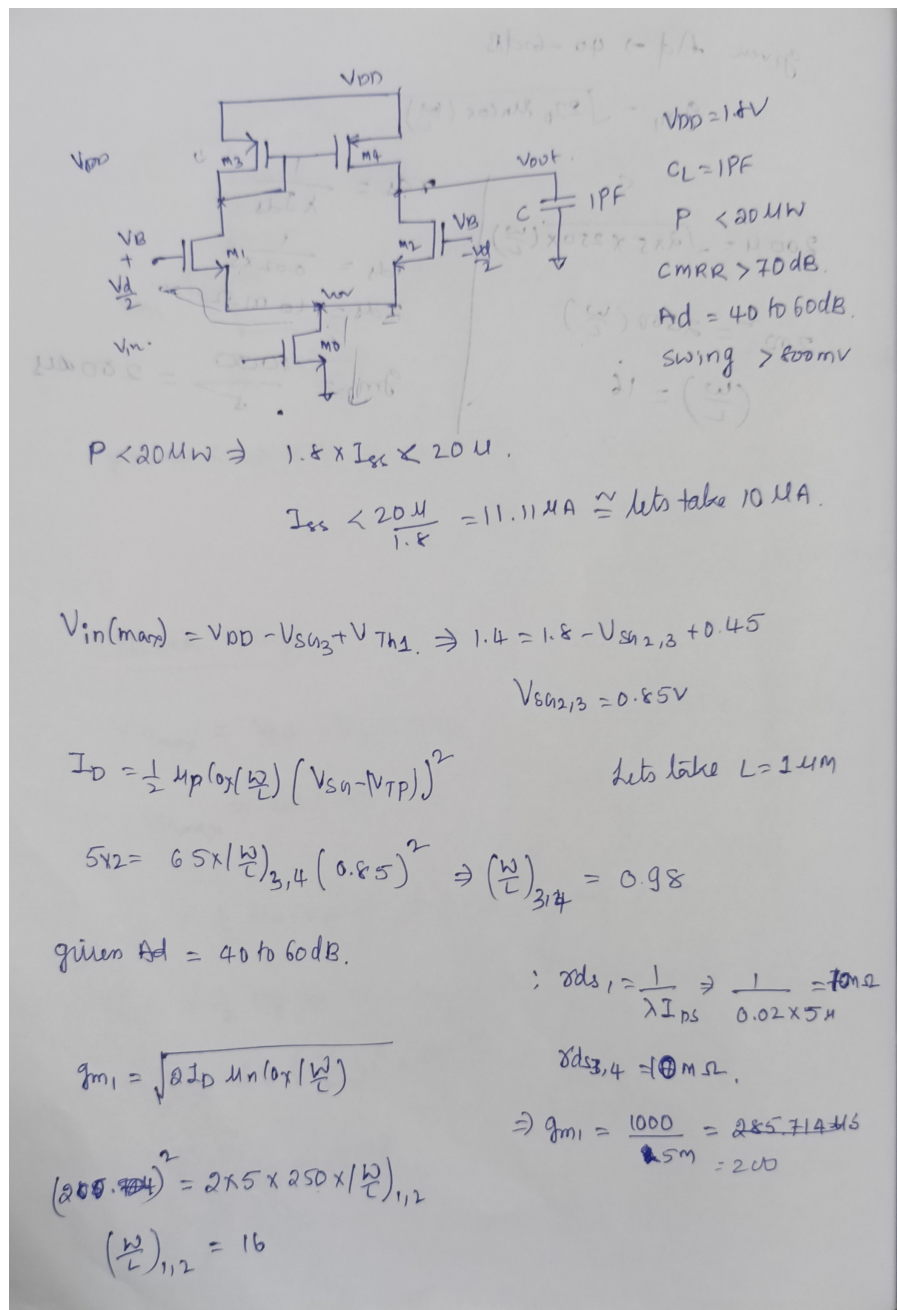


Figure 13: Analysis of OTA

$$\begin{aligned}
 V_{I(\min)} &= V_{DS1} + V_{DS0} \\
 &= V_{in} - V_{S1} + V_D \Rightarrow 0.8 = 0.65 + V_{DS} \\
 V_{DS} &= V_{OV} = 0.15 \text{ V.} \\
 \Rightarrow \text{not used } \sqrt{\frac{2I_D}{\mu_n \mu_p \left(\frac{W}{L}\right)}} + V_{TN} &= V_{S1} \\
 \Rightarrow \sqrt{\frac{10}{250 \times 16}} + 0.45 &= V_{S1} \Rightarrow V_{S1} = 0.5 \text{ V.} \\
 10 &= \frac{1}{2} \times 250 \times \left(\frac{W}{L}\right)_{\text{tail}} \times (0.15)^2 \quad \left\{ \begin{array}{l} r_{ds} = \frac{1}{\lambda I_{ds}} = \frac{1}{0.02 \times 524} \\ r_{ds} = 7 \text{ m}\Omega \\ g_{m1} = \frac{1000}{35} = 28 \end{array} \right. \\
 &= \frac{20}{250 \times 0.15^2} = 3.55.
 \end{aligned}$$

Figure 14: Analysis of OTA